

Lab 1: Exploratory Data Analysis

SDS3386 A00 - Data Science Lab

Fall 2026

Faculty of Science

University of Ottawa

Karina Glavan

Lab Date: Sept 16th 2026

Submission Date: Sept 21st 2026

Validation 1: Patients with heart disease are, on average older (mid-50s to 60s)

The data supports this validation. Patients without heart disease had a median age of 52 years, while patients with heart disease had a median age of 58 years. This indicates that patients diagnosed with heart disease tended to be older than those not diagnosed with heart disease.

This finding is further supported by the boxplot below.

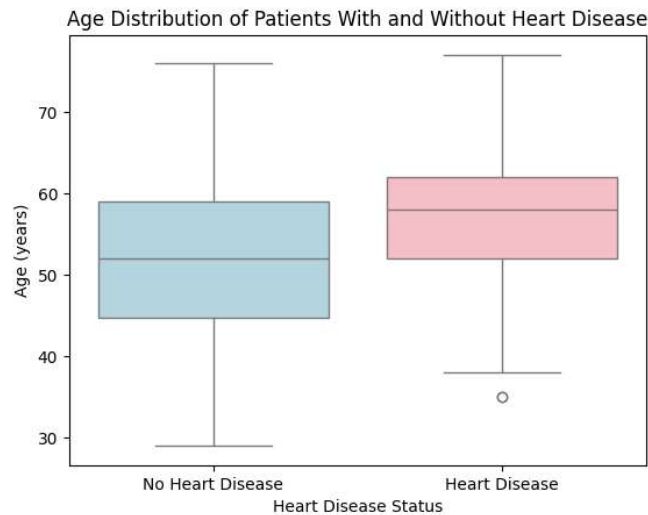


Figure 1: Age distribution of patients with and without heart disease.

The boxplot visually shows that the age distribution for patients with heart disease is shifted toward older ages, with a higher median than the group without heart disease.

Validation 2: Men are more represented in the dataset (around 68%), and more men than women are diagnosed

The number of men in the dataset is slightly more than double the number of women, with 206 men and 97 women. Men represent approximately 68% of the dataset, while women represent 32%. This demonstrates that men are substantially more represented in the dataset. The statement that more men than women are diagnosed is also supported by the data. Among male patients, 55.3% were diagnosed with heart disease, while among female patients, 25.8% were diagnosed.

Validation 3: Women with abnormal cholesterol or ECG results also show high prevalence

Women with either abnormal cholesterol or ECG results, or with both results being abnormal, show a higher prevalence of heart disease than women with neither abnormal finding. The prevalence among women with abnormal cholesterol is 27.7%, the prevalence among women with an abnormal ECG is 33.3%, and the prevalence among women with both is 34.1%. In comparison, the prevalence among women with neither abnormal finding is 10%.

Validation 4: Higher cholesterol and blood pressure are more common in patients with heart disease

First, combining patients with both high cholesterol and high blood pressure shows little difference between those with and without heart disease. Patients with both conditions and heart disease make up approximately 30% of the dataset, while those with both conditions and no heart disease make up approximately 29%. Plotting cholesterol against resting blood pressure in a scatterplot shows only a very weak positive correlation ($r = 0.13$).

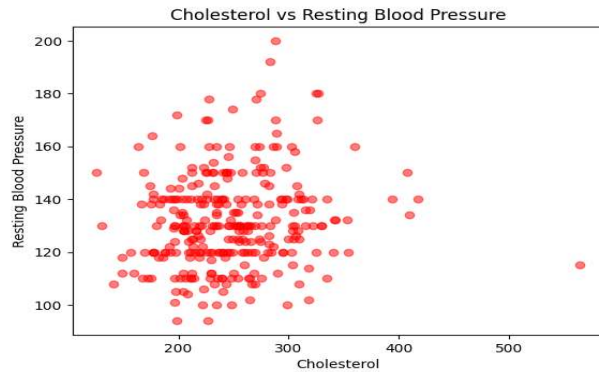


Figure 2: Relationship Between Cholesterol and Resting Blood Pressure.

Separately, the correlation coefficient between cholesterol and resting blood pressure was 0.13, demonstrating a very weak positive correlation between the two variables. However, when cholesterol and blood pressure are examined separately, a clearer difference is evident. The average cholesterol level of patients with heart disease was 251.47, compared with 242.64 for those without heart disease. Looking at **Figure 3**, patients with heart disease appear more concentrated within the higher cholesterol ranges than those without. The average resting blood pressure was also higher among patients with heart disease, at 134.57 compared with 129.25 for those without. **Figure 4** similarly shows a greater concentration of patients with heart disease toward the higher resting blood-pressure ranges.

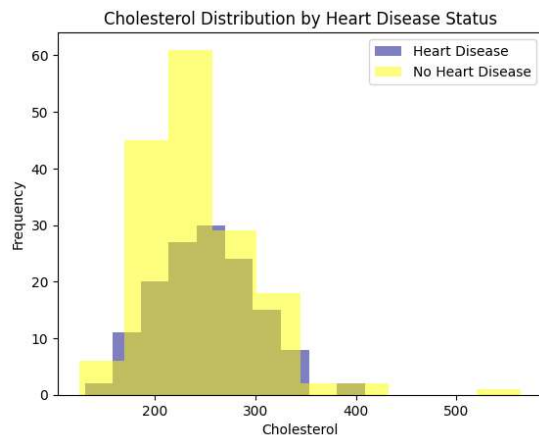


Figure 3: Cholesterol by Heart Disease Status

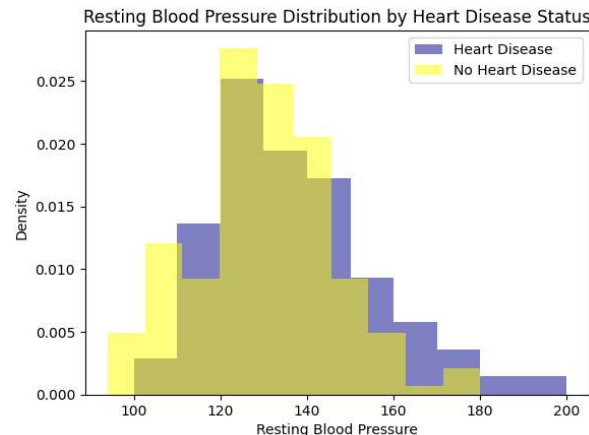


Figure 4: Resting Blood Pressure by Heart Disease Status

Validation 5: Patients with disease achieve lower maximum heart rates

This validation can be supported by comparing the average maximum heart rate of patients with heart disease and those without. The average maximum heart rate for patients with heart disease is 139.26, while the average for patients without heart disease is 158.38. These statistics clearly show that, on average, patients with heart disease achieve a lower maximum heart rate.

Validation 6: Patients that are classified as asymptomatic have a higher heart disease rate than patients that experience chest pain

The final validation can be supported by comparing the heart disease rate among asymptomatic patients with the rate among patients experiencing any form of chest pain. The heart disease rate among asymptomatic patients is 72.92%, while the rate among patients with chest pain is 21.38%. This clearly demonstrates a much higher heart disease rate among asymptomatic patients.

References

American Heart Association. (2025, August 14). *Understanding blood pressure readings*.

<https://www.heart.org/en/health-topics/high-blood-pressure/understanding-blood-pressure-readings>

Cleveland Clinic. (2025, October 13). *Cholesterol levels*.

<https://my.clevelandclinic.org/health/articles/11920-cholesterol-numbers-what-do-they-mean>

OpenAI. (2026). *ChatGPT (GPT-5.6 Sol)* [Large language model]. ChatGPT.

<https://chatgpt.com/>